

Ranking Training Data Quality

...

Group 7

Axel Riben, Felix Ölander, Marcus Bennevall
(and their trusty AI agents)

What's the problem?

LLM training runs are expensive, so we only want to train on high-quality data

But what is high quality?

- Hard to define
- Context-dependent
- Subjective



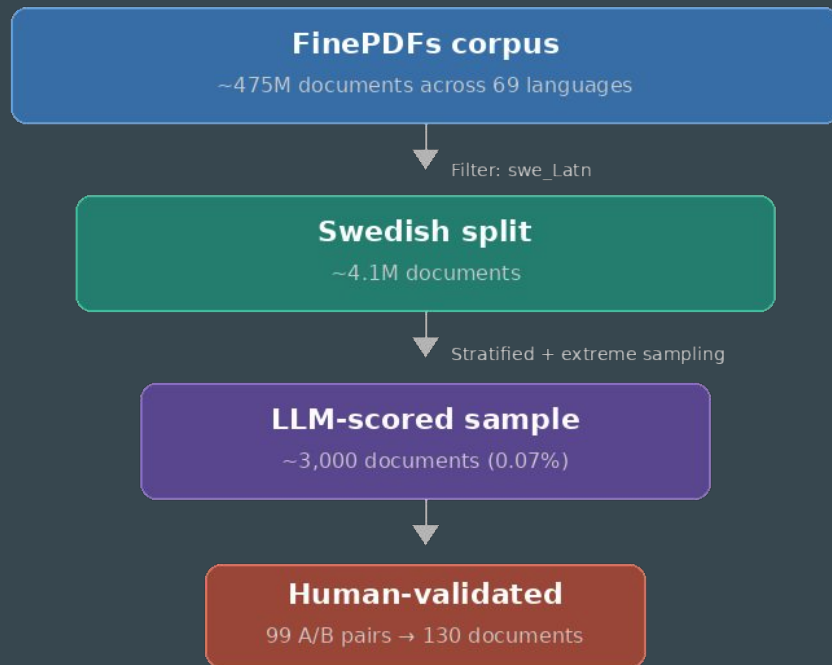
Research question and hypothesis

Research question : Which document-level signals best predict LLM pre-training usefulness in a Swedish web corpus?

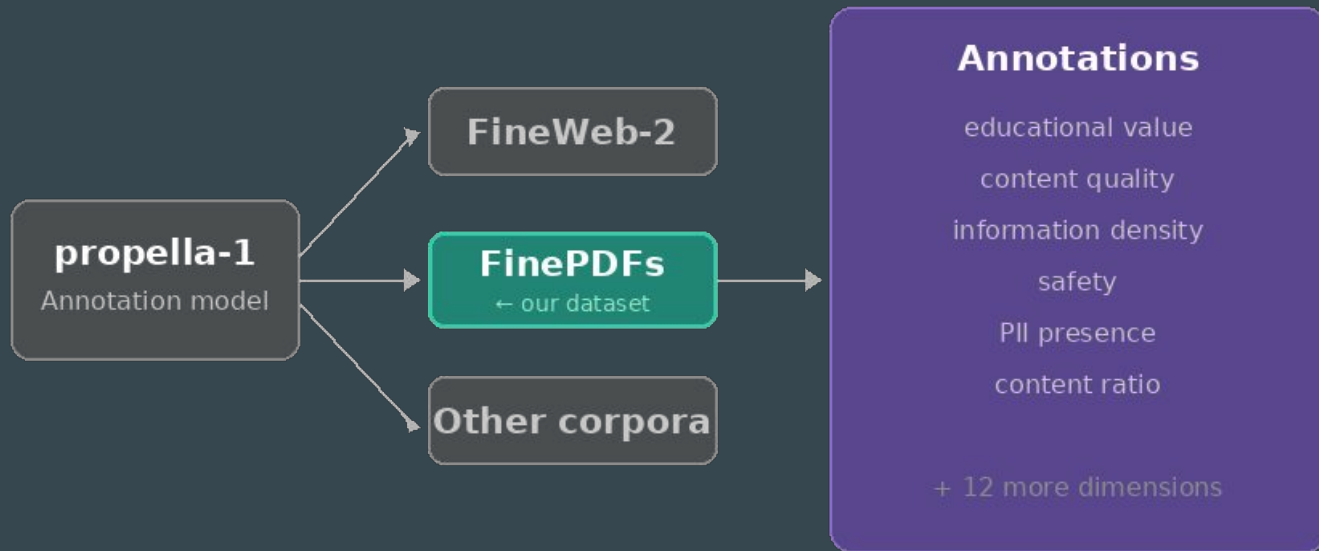
Hypothesis : A ranking model that combines decomposed quality signals – educational value, content quality, redundancy, and safety – will identify useful training documents more effectively than single-signal heuristic filters.

What data did we use?

FinePDFs



Propella



Independent from our LLM judge — separate annotation source

Human-scored data

263 Swedish documents drawn from FinePDFs

- Stratified across content quality x educational value
 - Content quality (“Overall writing and presentation quality”)
 - Educational value (“Potential for teaching and learning”)
- → 99 document pairs for human rating, with 3 constraints:
 - Same topic family
 - Similar length
 - A document can appear in max 5 pairs

Human rating: operationalizing the features

Content quality

- **Readability:** Which document is more readable and intact?
 - Is the text well-extracted? Coherent Swedish? Free from garbled characters, broken sentences, and extraction artifacts?

Educational value

- **Substance:** Which document contains more substantive, broadly useful information?
 - Does the content teach something widely applicable, or is it trivial, marketing copy or filler?

A/B test

45 pairs per rater

Two scrollable texts per pair

Two questions per pair

Document A

A collection of disconnected Swedish text fragments about architecture, history, and politics, likely from a download page or aggregated content site.

Hovslagargatan 4 : ett hus på Blasieholmen PDF EPUB LÄSA
ladda ner

Författare: Tulla Grünberger.

LADDA NER

LÄSA

Annan Information

Alla tre förslagen möjliggör besökare till Nobelcentret och till den nya parken för att komma fram med båt och gör det också möjligt att fortsätta förtöjning större fartyg, vilket därigenom bidrar till att bevara och kanske till och med stärka webbplatsens maritima karaktär. Designen ser bra ut på något sätt! Den här broderidesignen är lämplig för en 5x7-hoop: (om du behöver en annan storlek, vänligen kontakta mig! Broderade stygn är digitaliserade med broderade stygn och

Document B

Product brochure detailing specifications and features of various garage door opener models from Sommer and Aperto.

Garageportöppnare
Pålitligt, säkert och bekvämt

VI STÅR FÖR KVALITETEN – DU NJUTER AV BEKVÄMLIGHETEN!

En fjärrstyrd garageportöppnare ger komfort och trygghet för dig!

Garageportöppnare från SOMMER och APERTO sätter användarens krav i centrum: kvalitet, komfort och säkerhet har högsta prioritet.

Vi har modeller för alla typer av garageportar; takskjutportar, slagportar, in- och utåtgående vippportar samt sidosektionsportar.

Våra garageportöppnare går mycket tyst och mjukt och är underhållsfria, med lång livslängd.

Question 1 — Which document is more readable and intact?

A much more readable

A slightly more readable

B slightly more readable

B much more readable

Question 2 — Which document contains more substantive, broadly useful information?

A much more substantive

A slightly more substantive

B slightly more substantive

B much more substantive

Notes (optional, 200 char max)

Submit

Skip — this pair has extraction or language issues making it unreadable

Agreement and scoring

Human-propella agreement

- Content quality: 73.9%
- Educational value: 65.8%

Human inter-rater agreement 88.9% on both features

130 documents scored based on A/B testing

- Bradley-Terry model (wins vs losses)
- From 0.0 to 1.0
- (Theoretically) comparable to LLM-scored documents

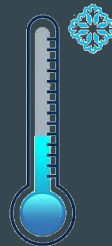
LLM-scored data

Model

- Gemma-4-31B-it (Berget AI, temperature 0)
- ~3,000 Swedish documents: ~2,000 stratified + ~1,000 targeted extremes

Scoring

- Independent scoring of the two axes educational value and content quality
- Why separate scoring? To avoid "halo effect": a well-written restaurant menu shouldn't score high just because it reads well
- Combined score: arithmetic mean $(x+y)/2$



Comparing LLM and Human scorings

- Content quality aligned with humans ($\rho = 0.41$). Educational value did not ($\rho = 0.18$). Combining them made things worse than cq alone ($\rho = 0.4$)
- Why? The prompt defined "educational" narrowly, scoring practical documents like fee tables and technical specs low, some of which humans preferred.
- Not just our prompt: even Propella's broader definition of educational value gets near-zero weight when trained on human preferences
- What the prompt tells LLM to measure will propagate through every downstream score

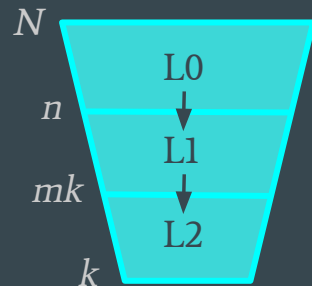
Cascading reranker model

Input: Propella and FinePDFs features (+ LLM-generated gold scores during training)

L0: Filtering and preprocessing

L1: Fast, coarse layer with simple(-ish) linear ranking

L2: Slow, non-linear layer that captures finer nuances



- **Gradient Boosted Decision Trees (GBDT)** : A series of experts (trees) that each corrects its predecessor's mistakes
- **Listwise Learning-to-Rank (LTR)** : Optimizes loss function for the *entire* ranking
- **Normalized Discounted Cumulative Gain @ k (NDCG@ k)**: Mistakes near the top hurt more

Evaluation and baselines

Test data: Human-scored data + held-out subset of LLM-scored data (10%)

Metrics: Spearman correlation (ρ) + NDCG@ k ($k \in \{10, 25, 50, 100\}$) + others

Baselines:

1. **Random ranking:** Hard bottom baseline
2. **Averaged input features:** Naive heuristic
3. **Educational_value > content_quality > safety > doc_id:** Simple heuristic #1
4. **Content_quality > educational_value > safety > doc_id:** Simple heuristic #2
5. **L1 ranking:** To measure the contribution of L2

Results

Human data (%)

LLM data (%)

	Spearman ρ	NDCG@10	NDCG@100	Spearman ρ	NDCG@10	NDCG@100
Random	0.3	13.5	51.0	-0.1	15.6	32.2
Average	25.3	24.8	63.9	80.1	39.1	75.7
Edu_val	19.4	29.1	62.2	81.1	60.5	83.6
Cont_qual	37.1	29.1	69.2	86.0	60.5	82.4
L1	27.8	24.8	64.8	84.4	40.1	77.0
L1 + L2	38.7	16.1	64.7	77.6	55.8	72.6

Research question and hypothesis

Research question : Which document-level signals best predict LLM pre-training usefulness in a Swedish web corpus?

Hypothesis : A ranking model that combines decomposed quality signals – educational value, content quality, redundancy, and safety – will identify useful training documents more effectively than single-signal heuristic filters.

Feature	L1 weight
Content quality	0.98
Information density	0.93
Educational value	0.89
Length	0.53
Language confidence	0.17
PII absence	0.12
Safety	0.00

Research question and hypothesis

Research question : Which document-level signals best predict LLM pre-training usefulness in a Swedish web corpus?

Hypothesis : A ranking model that combines decomposed quality signals – educational value, content quality, redundancy, and safety – will identify useful training documents more effectively than single-signal heuristic filters.

Feature	L1 weight
Content quality	0.98
Information density	0.93
Educational value	0.89
Length	0.53
Language confidence	0.17
PII absence	0.12
Safety	0.00

Limitations

Limited data:

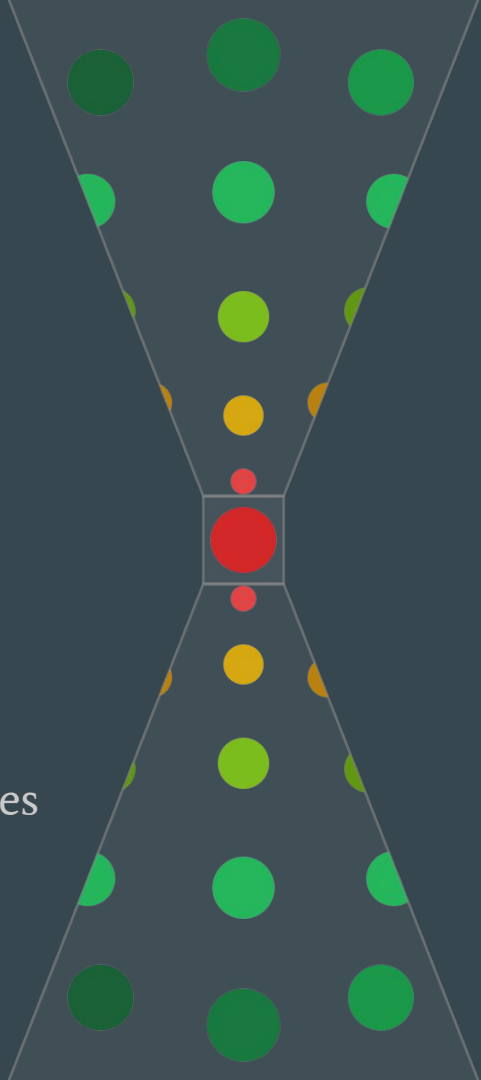
- Only tested Swedish
- Too little human data to accurately score each document
- Limited coordination between human and LLM data

Limited knowledge:

- Limited experience with LTR and GBDT models
- Flawed human judges

Operationalized features imperfectly represent underlying features

Quality doesn't necessarily correlate with some features



Takeaways and future research

Takeaways

- Simple heuristics are surprisingly good
- We need beefier machines
- Prompt design is critical
- Ironically difficult to produce good training data

Future research

- Actually use the reranker soft filter
- LLM A/B scoring

Thank you!

...